Programme

Tuesday, July 10, 2018; Academy Building*

16.00	Aula	*Introductory lecture: Sabine Hossenfelder* *Chair: F. A. Muller*		
17.30	Garden	*Welcome Drinks*		

Wednesday, July 11, 2018

	Session 1	Session 2	Session 3	Session 4
9.00	*Coffee and registration*			
9.30	**Pablo Acuña:** Kochen-Specker Theorem in the Context of von Neumann's 'Impossibility Proof'	**Natalja Deng:** Presentism, Triviality, and the Relativity Objection	**Alexander Blum:** Heisenberg's 1958 Weltformel and the roots of post-empirical physics	**Valeriya Chasova:** Observers, references and symmetries
10.10	**Gábor Hofer-Szabó:** What is quantum contextuality, and what is not?	**Cristian Lopez:** Time reversal, the arrow of time and metaphysical commitments	**Richard Dawid:** Finetuning and the Lack of Fundamental Free Parameters	**Pablo Ruiz de Olano:** Symmetries in Physics: Variational, Dynamical, and Hamiltonian
10.50	*Coffee Break*			
11.10	**Paul Busch**: Quantum measurement uncertainty: its origin and significance	**John Dougherty:** The inertial structure of a Newtonian cosmos	**Manuel Herrera:** Conservation laws: a philosophical analysis of their status	**Sebastien Rivat:** Renormalization Scrutinized
11.50	**Jos Uffink:** Schrödinger and the prehistory of the EPR argument.	**Tushar Menon:** Clocks and chronogeometry: Rotating spacetimes and the relativistic null hypothesis	**Taimara Passero:** The Geometrization as a Thema in the History of Physics	**Joshua Rosaler and Robert Harlander:** Naturalness, Wilsonian Renormalization, and "Fundamental Parameters" in Quantum Field Theory
13.30	*Plenary Lecture:* ***Emily Adlam*** *Chair: Fedde Benedictus*			
14.30	**Fabio Costa and Sally Shrapnel:** Contextuality in ontological models without causal assumptions	**David Wallace:** The Case for Black Hole Thermodynamics	**Anne Deng:** The Role of General Philosophy of Science in the Interpretation of Physical Theories: A Case Study	**James Fraser:** The Renormalization Scheme Dependence Problem: A Topography and Analysis
15.10	**Ronnie Hermens:** ψ-ontic models without ψ	**Jb Manchak:** Some "No Hole" Spacetime Properties Are Unstable	**Bryan W. Roberts:** The mechanics of markets	**Cristin Chall:** Model-Groups as Scientific Research Programmes
15.50	*Coffee Break*			
16.30	**Lev Vaidman:** Beyond "To be or not to be?" Degree and type of presence of a quantum particle in the past	**Elliott Chen:** On Maxwell Gravitation	**Thao Le and Alexandra Olaya-Castro:** Perceived objectivity via strong quantum Darwinism and spectrum broadcast structure	**Benjamin Feintzeig:** Quantization, Approximation, and Interpretation
17.10	**Ruward Mulder:** Emergence and Pragmatism in David Wallace's Emergent Multiverse	**J. Brian Pitts:** What Are Observables in Hamiltonian Einstein-Maxwell Theory?	**Radin Dardashti:** What Constitutes a Problem in Physics? The Case of the Strong CP Problem	**Olimpia Lombardi and Manuel Herrera:** Understanding decoherence by comparing classical and quantum irreversibility
[??]	*Public Lecture:* ***Sir Roger Penrose***			

Thursday, July 12, 2018

	Session 1 (zaal blauw)	Session 2	Session 3	Session 4
9.00	*Coffee and registration*			
9.30	**Raffael Krismer:** Pragmatist Quantum Mechanics	**Bixin Guo:** The Bare Manifold and the Metric Structure: The Nature of Spacetime in General Relativity	**Wayne Myrvold:** Explaining Thermodynamics: What Remains to be Done?	**Robert Bishop:** Determinism in Context
10.10	**Alexei Grinbaum:** What's in the input/output distinction?	**Adán Sus:** Symmetries, physical possibilities and spacetime	**Erik Curiel:** Irreversibility in Thermodynamics versus in Statistical Mechanics	**Marij van Strien:** Disentangling causality and determinism
10.50	*Coffee Break*			
11.10	**Sam Rijken:** Non-locality and timelike Bell-type inequalities	**Samuel Fletcher:** Reduction and Causal Set Theory's Hauptvermutung	**Valia Allori:** Some Remarks on Explanation in Statistical Mechanics	**Miklos Redei and Zalan Gyenis:** Categorial independence in categorial quantum field theory
11.50	**Howard Wiseman:** Three boos for locality	**Christian Wuthrich and Vincent Lam:** Laws beyond spacetime	**Aldo Filomeno:** Statistical necessity without a guiding dynamics	**Niels Linnemann and Kian Salimkhani:** Reassessing the Weinberg-Witten Theorem
13.30	*Plenary Lecture:* ***Charlotte Werndl*** *Chair: Guido Bacciagaluppi*			
14.30	**Oliver Reardon-Smith and Paul Busch:** Measurement uncertainty and covariance	**Daniel Sega Neuman:** Is Cosmic Inflation a Testable Theory?	**Patricia Palacios and Lapo Casetti:** Redefining Equilibrium for Long-range Interacting Systems	**Jeremy Butterfield:** On Dualities and Equivalences Between Physical Theories
15.10	**Leon Loveridge, Paul Busch and Takayuki Miyadera:** Symmetry, Reference Frames, and Relational Quantities in Quantum Mechanics	**Caspar Jacobs:** Why Cosmology Does Not Prove the Past Hypothesis	**Geoffrey Sewell:** Thermodynamic Completeness and the Differentiability of Entropy	**Sebastian De Haro:** Formulating Emergence in the Physical Sciences
15.50	*Coffee Break*			
16.30	**Leevi Leppäjärvi, Teiko Heinosaari and Sergey Filippov:** Measurement simulability in general probabilistic theories	**Juliusz Doboszewski:** On cosmological fatalism and large scale structure of spacetime	**Márton Gömöri:** Why do initial conditions in an actual sequence of experiments approximately follow the uniform distribution over phase space with respect to the Lebesgue measure?	**Niels Martens:** Symmetry-to-reality inferences: The Aharonov-Bohm Efffect as a Case Study for Motivational Realism
17.10	**Joanna Luc and Tomasz Placek:** On generalised probability space representations of quantum mechanical experiments	**Sylvia Wenmackers:** What the two-envelopes paradox teaches us about the two-headed arrow of time	**Joshua Luczak and Lena Zuchowski:** The Ehrenfest's Use of Toy Models to Explore Irreversibility in Statistical Mechanics	**Josh Hunt:** Symmetry and Degeneracy in the Hydrogen Atom
?18.30	*Dinner:* ***Ottone*****			

Friday, July 13, 2018

	Session 1 (zaal blauw)	Session 2	Session 3	Session 4
9.00	*Coffee and registration*			
9.30	**Paul M. Näger:** A Stronger Bell Argument for (Some Kind of) Parameter Dependence	**Mike D. Schneider:** Interpretation and crisis in the vacuum	**Sebastian Fortin, Hernán Accorinti and Jesus Alberto Jaimes Arriaga:** Phonons: a case of intra-theoretic relationship	**Nicolas Gisin:** Indeterminism in Physics, Real Numbers, Classical Chaos and Bohmian Mechanics
10.10	**Gijs Leegwater:** Turning indeterminism into determinism: Does it matter when God throws his dice?	**Karim Thebault and Sean Gryb:** Superpositions of the cosmological constant allow for singularity resolution and unitary evolution in quantum cosmology	**Stephan Eijt:** 1, 2, Many: Emergence in positron-electron systems	**Magdalena Zych, Fabio Costa, Igor Pikovski and Caslav Brukner:** Bell's Theorem for Temporal Order
10.50	*Coffee Break*			
11.10	**Tim Palmer:** A Finite Theory of Quantum Physics	**Rasmus Jaksland:** Probing spacetime with a holographic relation between spacetime and entanglement	**Jorge Alberto Manero Orozco:** Imprints of the Underlying Structure of Physical Theories	**C. D. McCoy:** Why is Planck's constant a universal constant?
11.50	**Siddhant Das and Detlef Duerr:** Arrival Time Distributions of Spin-1/2 Particles	**Alexander Smith:** Quantizing time: Interacting clocks and systems	**Angelika Mus-Nowak:** Is grounding metaphysics on the findings of current scientific theories justified?	**Peter Evans:** A sideways look at faithfulness
13.30	*Plenary Lecture:* ***Ted Jacobson*** *Chair: Dennis Dieks*			
14.30	**Maaneli Derakhshani:** Stochastic Mechanics is a Viable Foundation for Quantum Mechanics	**Luigi Laino:** Between Knots and Links: an Argument in Favour of the Transcendental Representation of Space in Loop Quantum Gravity	**Marco Giovanelli:** 'Like classical Thermodynamics before Boltzmann'. Why did Einstein Compare Relativity Theory with Thermodynamics?	**Jeremy Steeger:** Hidden Variable Theories and Subjective Probability
15.10	**Martin Ringbauer:** Experimental metaphysics: Probing the foundations of quantum theory	**Alexander Afriat:** Topology, holes & sources	**Vera Matarese:** Qantum Gravity: A Threat to Humeanism?	**Thomas De Saegher:** The Fate of Some Semantic Problems for Fuzzy Links in Relativistic Dynamical Reduction Models
15.50	*Coffee Break*			
16.30	**John van de Wetering:** Reconstruction of quantum theory from universal filters	**Kevin Kadowaki:** On Rotation Curve Analysis	**Vincent Ardourel:** Phase Transitions, Renormalization Group, and Finite Size Scaling Theory	**Timothy Schmitz:** Computing in GRW Quantum Mechanics
17.10	**Alexander Wilce:** Getting more out of the subspace axiom	**Simon Friederich:** Testing multiverse theories and the problem of researcher degrees of freedom	**Jingyi Wu:** Explaining Universality: Infinite Limit Systems in the Renormalization Group Method	**Antoine Tilloy:** A realist redefinition of interacting quantum field theories inspired from dynamical collapse models

Saturday, July 14, 2018

11.00	Excursion Kröller Müller Museum	Details TBA		
18.30				

*) City Centre, Domplein 29, Utrecht; Phone: 030 253 8252

**) Restaurant Ottone is located at Kromme Nieuwegracht 62, at 350 meters East of the Academy Building.